

Artificial Intelligence in Pediatric Radiology

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CHLA Radiology Grand Rounds

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Objectives

- Describe the big players in radiology AI, both general and pediatric
- Summarize what radiology AI does well and what remains unresolved

Methods

Journals, Preprints: PubMed, Embase, arXiv, medRxiv

Conferences: NeurIPS, ICLR, ICML, CVPR, MICCAI, MIDL

Datasets: Hand-curated list

Newsletters: The Imaging Wire, RSNA News, TLDR AI, TLDR Tech, Signify Research, ESR / ECR, Radiology Business

Commercial: FDA list of AI-enabled devices

Date range: January 1 2008 – September 30 2026

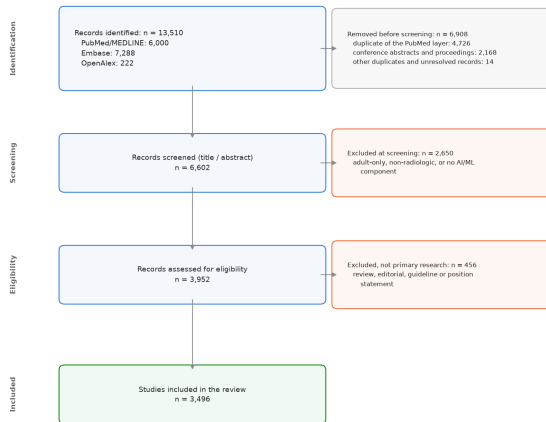
Search strategy: radiology AND AI [AND pediatric]

radiology: (radiology[tiab] OR radiological[tiab] OR radiograph*[tiab] OR "medical imaging"[tiab] OR "diagnostic imaging"[tiab] OR tomography[tiab] OR "magnetic resonance"[tiab] OR MRI[tiab] OR "MR imaging"[tiab] OR "MR images"[tiab] OR MRIs[tiab] OR "MR image"[tiab] OR "MR scan"[tiab] OR "MR scans"[tiab] OR "computed tomography"[tiab] OR CT[tiab] OR ultrasound[tiab] OR ultrasonograph*[tiab] OR sonograph*[tiab] OR echocardiograph*[tiab] OR mammograph*[tiab] OR "chest x-ray"[tiab] OR "x-ray"[tiab] OR fluoroscop*[tiab] OR scintigraph*[tiab] OR "positron emission"[tiab] OR PET[tiab] OR SPECT[tiab] OR angiograph*[tiab] OR neuroimaging[tiab] OR tractograph*[tiab] OR elastograph*[tiab] OR "fMRI"[tiab] OR "functional MRI"[tiab] OR "functional magnetic resonance"[tiab] OR connectome*[tiab] OR "diffusion tensor"[tiab] OR DTI[tiab] OR "bone age"[tiab] OR "skeletal age"[tiab] OR "skeletal maturity"[tiab] OR "barium enema"[tiab])

AI: ("artificial intelligence"[tiab] OR "machine learning"[tiab] OR "deep learning"[tiab] OR "convolutional neural network"[tiab] OR "convolutional neural networks"[tiab] OR "neural network"[tiab] OR "neural networks"[tiab] OR "computer-aided diagnosis"[tiab] OR "computer aided diagnosis"[tiab] OR "computer-aided detection"[tiab] OR radiomic*[tiab] OR "computer vision"[tiab] OR "large language model"[tiab] OR "large language models"[tiab] OR "foundation model"[tiab] OR "foundation models"[tiab] OR "transfer learning"[tiab] OR "vision transformer"[tiab] OR "self-supervised"[tiab] OR "random forest"[tiab] OR "support vector machine"[tiab] OR "gradient boosting"[tiab] OR "natural language processing"[tiab])

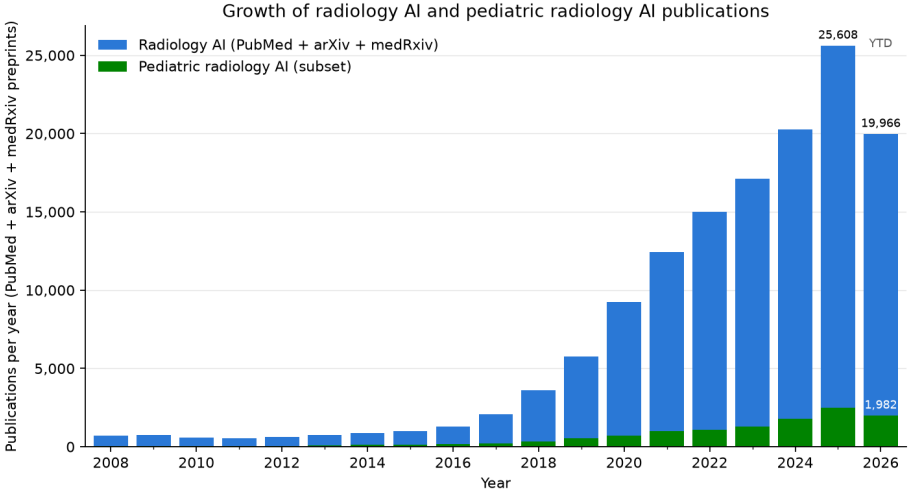
pediatric: (pediatric*[tiab] OR paediatric*[tiab] OR child*[tiab] OR infant*[tiab] OR neonat*[tiab] OR adolescen*[tiab] OR fetal[tiab] OR foetal[tiab] OR fetus*[tiab] OR foetus*[tiab] OR newborn*[tiab] OR preterm[tiab] OR "premature infant"[tiab] OR "premature infants"[tiab] OR "premature birth"[tiab] OR "premature neonate"[tiab] OR "premature neonates"[tiab] OR prenatal[tiab] OR antenatal[tiab] OR perinatal[tiab] OR youth[tiab] OR juvenile[tiab] OR "children's hospital"[tiab] OR schoolchild*[tiab] OR toddler*[tiab])

Study selection

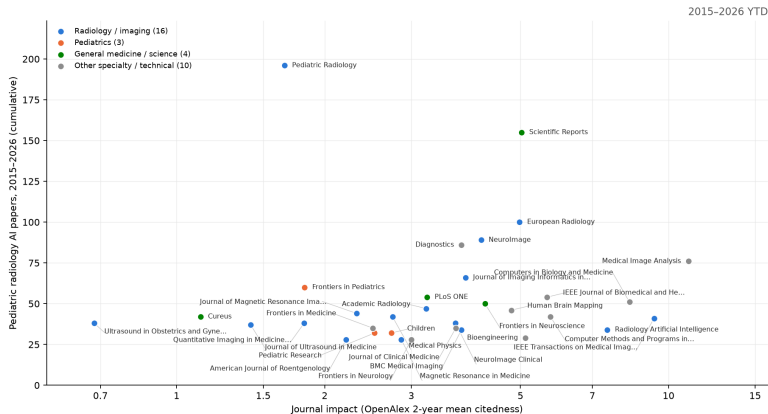


PubMed and Embase layers complete (2005-2026). Conference material is retrieved and reported but excluded by the publication-form criterion.

Radiology AI publication counts

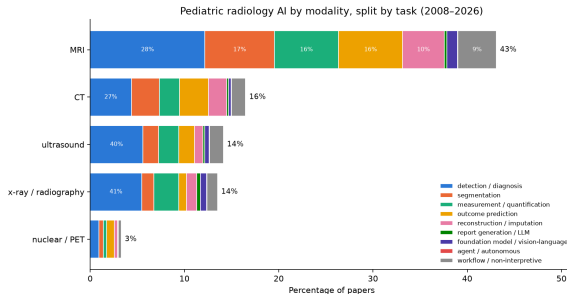


Which journals publish pediatric radiology AI



x = OpenAlex 2-year mean citedness (a free JIF analogue; runs below the published JIF). y = cumulative pediatric radiology-AI papers 2015-2026 (PubMed, strict title/abstract query). The 33 journals with at least 25 of them are shown, carrying 1,807 of 5,099; the rest is a long tail. Largest: Pediatric Radiology (196); median impact 3.7, 1 at 10 or above – this work lands mostly in mid-impact specialty and technical journals. A further 3 conference proceedings and preprint servers clear the threshold (94 papers) but have no impact number of any kind and are not plotted. The same chart animated year by year: [figures/journal_impact.gif](#).

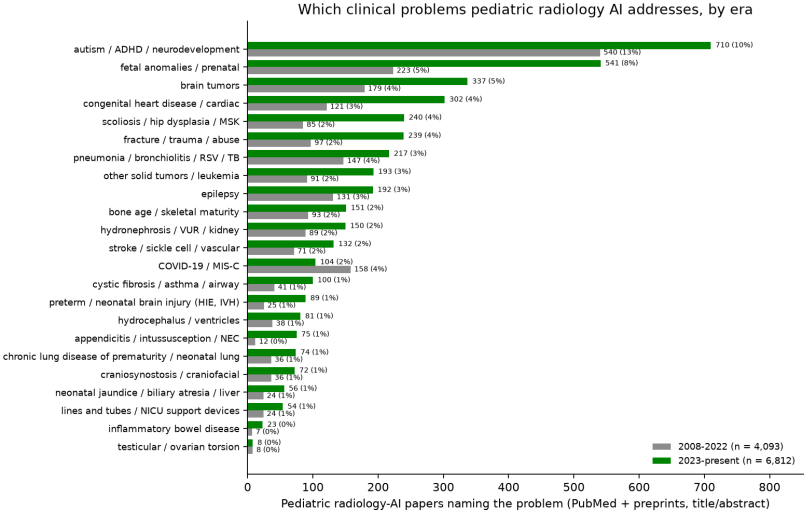
Pediatric radiology AI publications, stratified by modality and task



Same construction, restricted to the pediatric subset (10,737 papers incl. 596 preprints). Top pediatric modalities: MRI 41%, CT 16%, ultrasound 13%, x-ray / radiography 12%. Mammography is omitted: its 79 pediatric-query hits (of 10,737) are adult breast-imaging papers whose abstracts mention children.

- detection / diagnosis:** is a finding present now, and which one (label or box): fracture, pneumonia, tumor; includes triage and screening
- segmentation:** outline a structure or lesion (a mask)
- measurement / quantification:** a number from the image: bone age, Cobb angle, organ volume, fetal biometry
- outcome prediction:** a future risk, response, or survival estimate from the image (prognosis, radiomics signatures)
- reconstruction / imputation:** a better or missing image: lower dose, faster scans, denoising, synthetic CT/MR
- report generation / LLM:** text: draft, summarize, or extract from radiology reports
- foundation model / vision-language:** a large pretrained model reused across tasks (segment-anything, self-supervised)
- agent / autonomous:** multi-step actions taken without a human in the loop
- workflow / non-interpretive:** protocoling, scheduling, ordering, decision support, education

Pediatric radiology AI publications, stratified by medical problem



Most-cited radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	What it answers
14,326	1023.0	A survey on deep learning in medical image analysis (2017)	Medical Image Anal.	<i>survey</i> : What can deep learning do across medical imaging? (the field's reference survey)
5,658	446.6	Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation (2021)	ECCV Workshops	<i>segmentation</i>
2,990	340.4	Deep Learning in Medical Image Analysis (2017)	Annual Review of Biomedical Engineering	<i>survey</i> : How does deep learning apply to medical image analysis? (review)
2,553	290.0	Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images (2022)	BrainLes@MICCAI	<i>segmentation</i>
2,769	272.4	COVID-Net: a tailored deep convolutional neural network design for detection of COVID-19 cases from chest X-ray images (2020)	Scientific Reports	<i>classification</i> : Can a CNN detect COVID-19 on chest X-ray? (COVID-Net)
3,634	236.9	ChestX-Ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases (2017)	Computer Vision and Pattern Recognition	<i>classification</i>
10,044	226.3	nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation (2020)	Nature Methods	<i>segmentation</i> : Can one self-configuring method segment any organ or lesion without hand tuning? (nnU-Net)
4,100	170.2	Convolutional neural networks: an overview and application in radiology (2018)	Insights into Imaging	<i>tutorial</i> : What is a convolutional neural network, explained for radiologists?
5,587	158.4	Computational Radiomics System to Decode the Radiographic Phenotype (2017)	Cancer Research	<i>radiomics</i> : Can standardized image features (radiomics) be extracted reproducibly? (PyRadiomics)
3,322	149.0	Artificial intelligence in radiology (2018)	Nature Reviews. Cancer	<i>review</i> : How will AI change radiology practice? (Hosny et al.)

FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average)

Most-cited radiology AI papers, 2023

Cites	FWCI	Paper	Venue	What it answers
1,608	615.8	Segment anything in medical images	Nature Communications	<i>foundation</i> : Can a prompt-based 'segment anything' model work on medical images? (MedSAM)
2,099	243.8	Foundation models for generalist medical artificial intelligence	Nature	<i>foundation model</i> : What would a single generalist medical AI model that handles images, text and records look like? (perspective)
702	118.5	Redefining Radiology: A Review of Artificial Intelligence Integration in Medical Imaging	Diagnostics	<i>survey / review</i> : Where does AI fit in radiology workflow, from acquisition to reporting? (review)
509	75.2	Medical image analysis using deep learning algorithms	Frontiers in Public Health	<i>other</i>
562	74.5	A Study of CNN and Transfer Learning in Medical Imaging: Advantages, Challenges, Future Scope	Sustainability	<i>other</i>
861	69.6	Segment Anything Model for medical image analysis: an experimental study	Medical Image Anal.	<i>segmentation</i> : How well does the unmodified Segment Anything Model work on medical images? (it needs prompts and adaptation)
573	55.8	MRI-based brain tumor detection using convolutional deep learning methods and chosen machine learning techniques	BMC Medical Informatics and Decision Making	<i>oncology</i>
460	44.5	Brain Tumor Detection Based on Deep Learning Approaches and Magnetic Resonance Imaging	Cancers	<i>oncology</i>
391	42.2	SAM-Adapter: Adapting Segment Anything in Underperformed Scenes	2023 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)	<i>segmentation</i>
442	37.4	A generalist vision-language foundation model for diverse biomedical tasks	Nature Medicine	<i>other</i>

Most-cited radiology AI papers, 2024

Cites	FWCI	Paper	Venue	What it answers
1,074	225.2	VM-UNet: Vision Mamba UNet for Medical Image Segmentation	ACM Transactions on Multimedia Computing, Communications, and Applications (TOMCCAP)	<i>segmentation</i> : Does a state-space (Mamba) backbone beat convolutions and transformers for medical segmentation?
242	110.3	Generalist foundation models from a multimodal dataset for 3D computed tomography	Nature Biomedical Engineering	<i>other</i>
282	74.6	Artificial intelligence in neuro-oncology: advances and challenges in brain tumor diagnosis, prognosis, and precision treatment	npj Precision Oncology	<i>oncology</i>
279	72.6	METHodological RadiomIcs Score (METRICS): a quality scoring tool for radiomics research endorsed by EuSoMI	Insights into Imaging	<i>radiomics</i> : How should the quality of a radiomics study be scored? (METRICS checklist)
349	47.6	Deep learning for medical image segmentation: State-of-the-art advancements and challenges	Informatics in Medicine Unlocked	<i>segmentation</i>
317	43.4	Advances in Medical Image Segmentation: A Comprehensive Review of Traditional, Deep Learning and Hybrid Approaches	Bioengineering	<i>segmentation</i>
238	37.0	Employing deep learning and transfer learning for accurate brain tumor detection	Scientific Reports	<i>oncology</i>
242	36.0	Deep semi-supervised learning for medical image segmentation: A review	Expert systems with applications	<i>segmentation</i>
241	34.5	A review of deep learning-based information fusion techniques for multimodal medical image classification	Comput. Biol. Medicine	<i>survey / review</i>
298	31.8	Vision-language models for medical report generation and visual question answering: a review	Frontiers Artif. Intell.	<i>report generation / LLM</i> : What can vision-language models do for report generation and visual question answering? (review)

Most-cited radiology AI papers, 2025

Cites	FWCI	Paper	Venue	What it answers
258	225.8	Deep Convolutional Neural Networks in Medical Image Analysis: A Review	Inf.	<i>survey / review</i>
145	158.8	Multi-model assurance analysis showing large language models are highly vulnerable to adversarial hallucination attacks during clinical decision support	Communications Medicine	<i>LLM / reports</i>
122	108.2	Multimodal generative AI for medical image interpretation	Nature	<i>other</i>
313	94.9	Towards generalist foundation model for radiology by leveraging web-scale 2D&3D medical data	Nature Communications	<i>foundation model</i> : Can a radiology foundation model trained on web-scale 2D and 3D scans generalize across modalities and tasks? (RadFM)
173	87.2	Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology	Nature Cancer	<i>agent / autonomous</i> : Can an autonomous LLM agent that calls imaging and other tools support oncology decisions? (validation study)
208	65.9	A review of deep learning for brain tumor analysis in MRI	npj Precision Oncology	<i>survey / review</i>
150	65.8	Med-R1: Reinforcement Learning for Generalizable Medical Reasoning in Vision-Language Models	IEEE Transactions on Medical Imaging	<i>LLM / reports</i>
233	64.6	Artificial intelligence in healthcare and medicine: clinical applications, therapeutic advances, and future perspectives	European Journal of Medical Research	<i>other</i>
151	54.0	Medical Image Segmentation: A Comprehensive Review of Deep Learning-Based Methods	Tomography	<i>segmentation</i>
118	39.5	Advanced Brain Tumor Classification in MR Images Using Transfer Learning and Pre-Trained Deep CNN Models	Cancers	<i>oncology</i>

Most-cited radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	What it answers
48	102.6	A generalizable foundation model for analysis of human brain MRI	Nature Neuroscience	<i>other</i>
16	90.5	Reporting checklist for foundation and large language models in medical research (REFINE): an international consensus guideline.	Diagnostic and Interventional Radiology	<i>LLM / reports</i>
15	76.1	Scaling medical AI across clinical contexts	Nature Medicine	<i>foundation model</i> : How can medical AI be scaled across hospitals, populations and tasks without breaking? (perspective)
27	67.3	Applications and limitations of large language models to integrate medical context: a comprehensive review	Iran Journal of Computer Science	<i>survey / review</i>
18	62.7	A hybrid deep learning framework based on VGG19 and U-Net for accurate brain tumor segmentation in MRI images	Magnetic Resonance Letters	<i>segmentation</i>
37	58.8	Explainable artificial intelligence (XAI) in medical imaging: a systematic review of techniques, applications, and challenges	BMC Medical Imaging	<i>survey / review</i> : Which explainability methods are used in medical imaging AI and do they help? (review)
12	45.3	Explainable AI-driven MRI-based brain tumor classification: a novel deep learning approach	Frontiers Artif. Intell.	<i>oncology</i>
21	44.2	Agentic AI in Radiology: Evolution from Large Language Models to Future Clinical Integration.	Radiology: Artificial Intelligence	<i>agent / autonomous</i> : What are agentic AI systems in radiology and what stands between them and clinical use? (review)
17	28.4	Leveraging multi-modal foundation models for analysing spatial multi-omic and histopathology data	Nature Biomedical Engineering	<i>other</i>
19	21.4	BIBSNet: A deep learning baby image brain segmentation network for MRI scans	Developmental Cognitive Neuroscience	<i>segmentation</i> : Can one network segment infant brain MRI across the first years of life? (BIBSNet)

Most-cited pediatric radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	What it answers
426	328.6	Performance of a Deep-Learning Neural Network Model in Assessing Skeletal Maturity on Pediatric Hand Radiographs. (2017)	Radiology	<i>bone age</i> : Can a CNN assess skeletal maturity as accurately as radiologists?
431	321.5	Fully Automated Deep Learning System for Bone Age Assessment (2017)	Journal of digital imaging	<i>bone age</i> : Fully automated bone age from hand radiographs
403	221.2	Deep learning for automated skeletal bone age assessment in X-ray images (2017)	Medical Image Anal.	<i>bone age</i> : Deep learning for automated bone age (BoNet)
143	73.0	Computerized Bone Age Estimation Using Deep Learning Based Program: Evaluation of the Accuracy and Efficiency. (2017)	AJR. American journal of roentgenology	<i>bone age</i> : Is a commercial deep-learning bone-age program accurate and faster in practice?
191	57.8	A Review on Deep-Learning Algorithms for Fetal Ultrasound-Image Analysis (2022)	Medical Image Anal.	<i>survey / review</i>
352	57.4	Standard Plane Localization in Fetal Ultrasound via Domain Transferred Deep Neural Networks (2015)	IEEE journal of biomedical and health informatics	<i>fetal MRI</i>
194	57.0	Automatic Fetal Ultrasound Standard Plane Recognition Based on Deep Learning and IIoT (2021)	IEEE Transactions on Industrial Informatics	<i>fetal MRI</i>
233	30.3	Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes (2020)	Scientific Reports	<i>fetal MRI</i>
458	27.0	The RSNA Pediatric Bone Age Machine Learning Challenge. (2019)	Radiology	<i>bone age</i> : How well can crowdsourced AI estimate bone age? (RSNA challenge, 260 teams)
253	25.8	An ensemble of neural networks provides expert-level prenatal detection of complex congenital heart disease (2021)	Nature Medicine	<i>detection</i>

Most-cited pediatric radiology AI papers, 2023

Cites	FWCI	Paper	Venue	What it answers
32	36.6	Bone Age Assessment Using Artificial Intelligence in Korean Pediatric Population: A Comparison of Deep-Learning Models Trained With Healthy Chronological and Greulich-Pyle Ages as Labels	Korean Journal of Radiology	<i>bone age</i>
62	33.6	Standard fetal ultrasound plane classification based on stacked ensemble of deep learning models	Expert systems with applications	<i>fetal ultrasound</i> : Can an ensemble of networks classify the standard fetal ultrasound planes?
27	33.5	Bone age assessment based on deep neural networks with annotation-free cascaded critical bone region extraction	Frontiers in Artificial Intelligence	<i>bone age</i>
43	23.8	Review on deep learning fetal brain segmentation from Magnetic Resonance images	Artif. Intell. Medicine	<i>segmentation</i>
37	20.2	Transfer learning for accurate fetal organ classification from ultrasound images: a potential tool for maternal healthcare providers	Scientific Reports	<i>fetal MRI</i>
34	16.6	Deep Learning-Based Multiclass Brain Tissue Segmentation in Fetal MRIs	Italian National Conference on Sensors	<i>segmentation</i>
43	9.7	Deeplasia: deep learning for bone age assessment validated on skeletal dysplasias	Pediatric Radiology	<i>bone age</i>
30	7.8	Deep learning for estimation of fetal weight throughout the pregnancy from fetal abdominal ultrasound.	American Journal of Obstetrics & Gynecology MFM	<i>fetal MRI</i>
38	7.3	Deep learning model for prenatal congenital heart disease (CHD) screening can be applied to retrospective imaging from the community setting, outperforming initial clinical detection in a well-annotated cohort	Ultrasound in Obstetrics and Gynecology	<i>fetal ultrasound</i> : Does a prenatal congenital heart disease screening model work on retrospective real-world scans?
31	5.8	MRI-Based End-To-End Pediatric Low-Grade Glioma Segmentation and Classification	Canadian Association of Radiologists journal = Journal l'Association canadienne des radiologistes	<i>segmentation</i>

Most-cited pediatric radiology AI papers, 2024

Cites	FWCI	Paper	Venue	What it answers
17	34.8	Intrapartum Ultrasound Image Segmentation of Pubic Symphysis and Fetal Head Using Dual Student-Teacher Framework with CNN-VIT Collaborative Learning	International Conference on Medical Image Computing and Computer-Assisted Intervention	<i>segmentation</i>
30	23.8	FetalBrainAwareNet: Bridging GANs with anatomical insight for fetal ultrasound brain plane synthesis	Comput. Medical Imaging Graph.	<i>fetal MRI</i>
20	19.5	Artificial intelligence assisted common maternal fetal planes prediction from ultrasound images based on information fusion of customized convolutional neural networks	Frontiers in Medicine	<i>fetal MRI</i>
18	15.9	Deep learning denoising reconstruction for improved image quality in fetal cardiac cine MRI	Frontiers in Cardiovascular Medicine	<i>reconstruction</i> : Does deep-learning denoising improve fetal cardiac cine MRI?
30	15.1	RTSeg-net: A lightweight network for real-time segmentation of fetal head and pubic symphysis from intrapartum ultrasound images	Comput. Biol. Medicine	<i>segmentation</i>
16	13.4	FetSAM: Advanced Segmentation Techniques for Fetal Head Biometrics in Ultrasound Imagery	IEEE Open Journal of Engineering in Medicine and Biology	<i>segmentation</i>
23	7.0	Deep learning-based automated lesion segmentation on pediatric focal cortical dysplasia II preoperative MRI: a reliable approach	Insights into Imaging	<i>epilepsy</i> : Can focal cortical dysplasia lesions be found automatically on preoperative pediatric MRI?
18	5.3	Integration of a Deep Convolutional Neural Network With Adaptive Channel Weight Technique for Automated Identification of Standard Fetal Biometry Planes	IEEE Transactions on Instrumentation and Measurement	<i>fetal MRI</i>
18	5.2	Explainable AI in Deep Learning Based Classification of Fetal Ultrasound Image Planes	Procedia Computer Science	<i>fetal MRI</i>
19	4.8	A deep learning framework for identifying and segmenting three vessels in fetal heart ultrasound images	BioMedical Engineering OnLine	<i>segmentation</i>

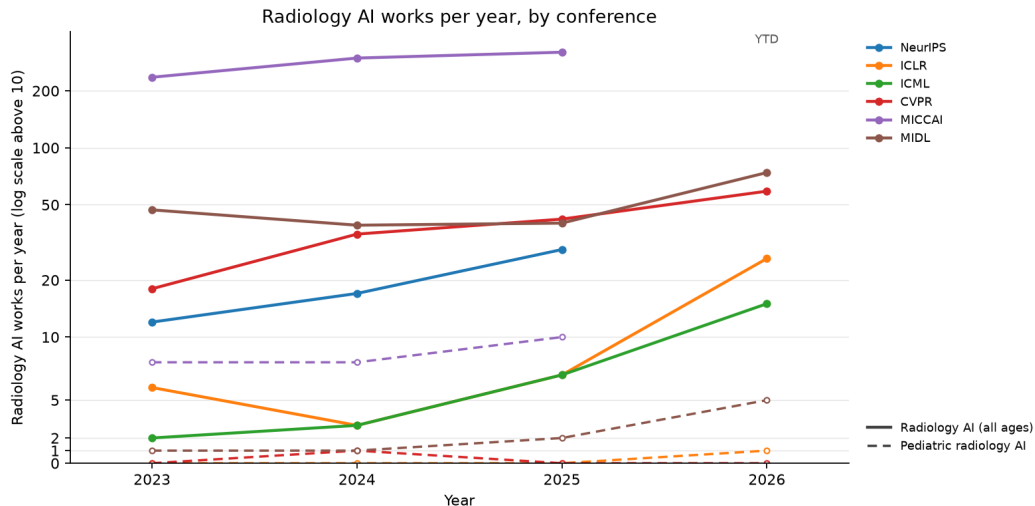
Most-cited pediatric radiology AI papers, 2025

Cites	FWCI	Paper	Venue	What it answers
16	171.3	A comprehensive review of artificial intelligence - based algorithm towards fetal facial anomalies detection (2013–2024)	Artificial Intelligence Review	<i>survey / review</i>
18	85.1	M-CNN-RF: A hybrid deep learning model for accurate pediatric skeletal age estimation using hand bone radiographs	Alexandria Engineering Journal	<i>other</i>
20	77.0	Bone Age Assessment Using Various Medical Imaging Techniques Enhanced by Artificial Intelligence	Diagnostics	<i>bone age</i>
18	24.9	Advancing prenatal healthcare by explainable AI enhanced fetal ultrasound image segmentation using U-Net++ with attention mechanisms	Scientific Reports	<i>segmentation</i>
12	21.8	FetalMovNet: A Novel Deep Learning Model Based on Attention Mechanism for Fetal Movement Classification in US	IEEE Access	<i>fetal MRI</i>
11	14.5	Effectiveness and clinical impact of using deep learning for first-trimester fetal ultrasound image quality auditing	BMC Pregnancy and Childbirth	<i>fetal MRI</i>
12	12.4	Association between deep learning radiomics based on placental MRI and preeclampsia with fetal growth restriction: A multicenter study.	European Journal of Radiology	<i>radiomics</i>
16	8.7	Prenatal detection of congenital heart defects using the deep learning-based image and video analysis: protocol for Clinical Artificial Intelligence in Fetal Echocardiography (CAIFE), an international multicentre multidisciplinary study	BMJ Open	<i>fetal ultrasound: Can deep learning on prenatal ultrasound video detect congenital heart defects?</i>
11	8.6	Multi-view deep learning framework for the detection of chest X-rays compatible with pediatric pulmonary tuberculosis	Nature Communications	<i>detection</i>
11	7.0	Evaluation of Image Quality and Scan Time Efficiency in Accelerated 3D T1-Weighted Pediatric Brain MRI Using Deep Learning-Based Reconstruction	Korean Journal of Radiology	<i>reconstruction</i>

Most-cited pediatric radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	What it answers
11	34.3	Artificial intelligence and radiomics for pediatric brain tumor classification and molecular characterization: a systematic review	Neuroradiology	<i>oncology</i> : Can AI and radiomics classify pediatric brain tumors and their molecular subtypes? (review)
5	27.9	Deep learning assessment of fetal brain maturation on 3D ultrasound volumes in early-onset fetal growth restriction	Ultrasound in Obstetrics and Gynecology	<i>fetal ultrasound</i> : Can deep learning grade fetal brain maturation on 3D ultrasound in growth restriction?
3	17.6	Foundation models in radiology: a primer for pediatric radiologists	Pediatric Radiology	<i>foundation model</i> : What should a pediatric radiologist know about foundation models? (primer)
2	17.6	Deep Learning-Based Multi-Class Pediatric Wrist Fracture Subtype Classification: A Pilot Study Comparing Convolutional Neural Network Architectures	Journal of Imaging	<i>fracture</i>
3	17.3	Current applications and challenges of artificial intelligence applied to diagnostics in pediatric musculoskeletal imaging	Pediatric Radiology	<i>classification</i>
4	13.6	Data mining in pediatric radiology in the era of artificial intelligence.	Pediatric Radiology	<i>other</i>
2	13.2	Acquisition time/dose reduction in pediatric PET imaging using patch-based deep learning	EJNMMI Physics	<i>reconstruction</i> : Can pediatric PET use one fifth of the counts with patch-based deep-learning denoising?
2	12.6	Optimizing pediatric chest CT: superior image quality and lower radiation dose with deep learning reconstruction.	European Journal of Radiology	<i>reconstruction</i> : Can deep-learning reconstruction lower pediatric chest CT dose while improving image quality?
2	12.2	Enhancing efficiency in pediatric brain tumor segmentation using a pathologically diverse single-center clinical dataset	Neuro-Oncology Advances	<i>segmentation</i>
3	11.7	Automated bone age assessment in rare pediatric growth disorders: a comparative study using Deeplasia	Frontiers in Endocrinology	<i>bone age</i> : Does an open bone-age model hold up in rare growth disorders against several experts? (Deeplasia)

Radiology AI works per year, by conference



NeurIPS: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	24	–	3D-RAD: A Comprehensive 3D Radiology Med-VQA Dataset with Multi-Temporal Analysis and Diverse Diagnostic Tasks
2025	20	–	Better Tokens for Better 3D: Advancing Vision-Language Modeling in 3D Medical Imaging
2024	47	2.0	Touchstone Benchmark: Are We on the Right Way for Evaluating AI Algorithms for Medical Segmentation?
2024	26	0.9	Eye-gaze Guided Multi-modal Alignment for Medical Representation Learning
2024	25	1.2	Enhancing vision-language models for medical imaging: bridging the 3D gap with innovative slice selection
2024	20	–	DiffusionBlend: Learning 3D Image Prior through Position-aware Diffusion Score Blending for 3D Computed Tomography Reconstruction
2023	144	–	SegVol: Universal and Interactive Volumetric Medical Image Segmentation
2023	104	–	LVM-Med: Learning Large-Scale Self-Supervised Vision Models for Medical Imaging via Second-order Graph Matching
2023	72	–	EHRXQA: A Multi-Modal Question Answering Dataset for Electronic Health Records with Chest X-ray Images
2023	66	–	SwiFT: Swin 4D fMRI Transformer
2023	64	–	NeuroGraph: Benchmarks for Graph Machine Learning in Brain Connectomics
2023	19	–	Knowledge-based in silico models and dataset for the comparative evaluation of mammography AI for a range of breast characteristics, lesion conspicuities and doses

ICLR: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	70	–	Large-scale and Fine-grained Vision-language Pre-training for Enhanced CT Image Understanding
2025	13	–	SimXRD-4M: Big Simulated X-ray Diffraction Data and Crystal Symmetry Classification Benchmark
2025	9	–	Self-Supervised Diffusion MRI Denoising via Iterative and Stable Refinement
2025	9	–	Score-based Self-supervised MRI Denoising
2024	143	–	MMed-RAG: Versatile Multimodal RAG System for Medical Vision Language Models
2024	31	–	MediConfusion: Can you trust your AI radiologist? Probing the reliability of multimodal medical foundation models
2024	19	–	The Effect of Intrinsic Dataset Properties on Generalization: Unraveling Learning Differences Between Natural and Medical Images
2024	16	–	LeFusion: Controllable Pathology Synthesis via Lesion-Focused Diffusion Models
2023	107	–	Advancing Radiograph Representation Learning with Masked Record Modeling
2023	84	–	DDM2: Self-Supervised Diffusion MRI Denoising with Generative Diffusion Models
2023	40	–	AE-FLOW: Autoencoders with Normalizing Flows for Medical Images Anomaly Detection
2023	18	–	Improved Training of Physics-Informed Neural Networks Using Energy-Based Priors: a Study on Electrical Impedance Tomography
2026	–	–	Fetal-Gauge: A Benchmark For Assessing Vision-Language Models In Fetal Ultrasound ★

ICML: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	21	–	CLIMB: Data Foundations for Large Scale Multimodal Clinical Foundation Models
2025	17	–	MindLLM: A Subject-Agnostic and Versatile Model for fMRI-to-Text Decoding
2025	11	–	Raptor: Scalable Train-Free Embeddings for 3D Medical Volumes Leveraging Pretrained 2D Foundation Models
2025	5	–	Efficient Noise Calculation in Deep Learning-based MRI Reconstructions
2025	2	–	LangDAug: Langevin Data Augmentation for Multi-Source Domain Generalization in Medical Image Segmentation
2025	0	–	Staged and Physics-Grounded Learning Framework with Hyperintensity Prior for Pre-Contrast MRI Synthesis
2024	14	–	Unsupervised Domain Adaptation for Anatomical Structure Detection in Ultrasound Images
2024	10	–	Adaptive Sampling of k-Space in Magnetic Resonance for Rapid Pathology Prediction
2023	17	–	ACAT: Adversarial Counterfactual Attention for Classification and Detection in Medical Imaging
2023	16	–	Robustness of Deep Learning for Accelerated MRI: Benefits of Diverse Training Data
2023	15	–	Learning High-Order Relationships of Brain Regions
2023	8	–	Contextualized Policy Recovery: Modeling and Interpreting Medical Decisions with Adaptive Imitation Learning

CVPR: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	68	265.4	Enhanced Contrastive Learning with Multi-view Longitudinal Data for Chest X-ray Report Generation
2024	84	47.9	Towards Generalizable Tumor Synthesis
2024	81	14.6	Rethinking Diffusion Model for Multi-Contrast MRI Super-Resolution
2024	73	13.8	MemSAM: Taming Segment Anything Model for Echocardiography Video Segmentation
2023	243	20.2	Dynamic Graph Enhanced Contrastive Learning for Chest X-Ray Report Generation
2023	222	75.5	Ambiguous Medical Image Segmentation Using Diffusion Models
2023	217	18.7	METransformer: Radiology Report Generation by Transformer with Multiple Learnable Expert Tokens
2023	164	21.9	MCF: Mutual Correction Framework for Semi-Supervised Medical Image Segmentation
2023	133	14.4	KiUT: Knowledge-injected U-Transformer for Radiology Report Generation
2023	117	67.2	Structure-Aware Sparse-View X-Ray 3D Reconstruction
2023	103	11.0	Label-Free Liver Tumor Segmentation
2023	75	14.1	Learning Federated Visual Prompt in Null Space for MRI Reconstruction
2024	22	–	M3-UDA: A New Benchmark for Unsupervised Domain Adaptive Fetal Cardiac Structure Detection ★

MICCAI: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	208	1.1	MedVLM-R1: Incentivizing Medical Reasoning Capability of Vision-Language Models (VLMs) via Reinforcement Learning
2024	133	–	CT2Rep: Automated Radiology Report Generation for 3D Medical Imaging
2024	102	–	Anatomically-Controllable Medical Image Generation with Segmentation-Guided Diffusion Models
2024	87	–	MedCLIP-SAM: Bridging Text and Image Towards Universal Medical Image Segmentation
2023	462	–	MedNeXt: Transformer-driven Scaling of ConvNets for Medical Image Segmentation
2023	128	–	CoLa-Diff: Conditional Latent Diffusion Model for Multi-Modal MRI Synthesis
2023	105	–	DiffMIC: Dual-Guidance Diffusion Network for Medical Image Classification
2023	97	13.9	Beyond Adapting SAM: Towards End-to-End Ultrasound Image Segmentation via Auto Prompting
2023	95	–	Self-Supervised MRI Reconstruction with Unrolled Diffusion Models
2023	83	–	Make-A-Volume: Leveraging Latent Diffusion Models for Cross-Modality 3D Brain MRI Synthesis
2023	72	19.0	BerDiff: Conditional Bernoulli Diffusion Model for Medical Image Segmentation
2023	71	–	Ariadne's Thread: Using Text Prompts to Improve Segmentation of Infected Areas from Chest X-ray images

MICCAI: pediatric radiology AI works, 2024–2026

Year	Population	Paper
2025	Fetal	Accurate and Efficient Fetal Birth Weight Estimation from 3D Ultrasound
2025	Fetal	Adaptive Frame Selection for Gestational Age Estimation from Blind Sweep Fetal Ultrasound Videos
2025	Infant / neonatal	Flexibly Distilled 3D Rectified Flow with Anatomical Constraints for Developmental Infant Brain MRI Prediction
2025	Fetal	Fusing Radiomic Features with Deep Representations for Gestational Age Estimation in Fetal Ultrasound Images
2025	Fetal	General Methods Make Great Domain-specific Foundation Models: A Case-study on Fetal Ultrasound
2025	Fetal	Influence of Classification Task and Distribution Shift Type on OOD Detection in Fetal Ultrasound
2025	Fetal	Latent Motion Profiling for Annotation-free Cardiac Phase Detection in Adult and Fetal Echocardiography Videos
2025	Fetal	Robust Fetal Pose Estimation across Gestational Ages via Cross-Population Augmentation
2025	Fetal	Self-supervised Normality Learning and Divergence Vector-guided Model Merging for Zero-shot Congenital Heart Disease Detection in Fetal Ultrasound Videos
2025	Fetal	You Can Detect It: Fetal Biometric Estimation Using Ellipse Detection
2024	Fetal	Cycle-consistent Learning for Fetal Cortical Surface Reconstruction
2024	Pediatric	Efficient and Gender-adaptive Graph Vision Mamba for Pediatric Bone Age Assessment
2024	Fetal	Fetal MRI Reconstruction by Global Diffusion and Consistent Implicit Representation
2024	Fetal	Geometric Transformation Uncertainty for Improving 3D Fetal Brain Pose Prediction from Freehand 2D Ultrasound Videos
2024	Fetal	Improving cross-domain brain tissue segmentation in fetal MRI with synthetic data
2024	Fetal	Intrapartum Ultrasound Image Segmentation of Pubic Symphysis and Fetal Head Using Dual Student-Teacher Framework with CNN-ViT Collaborative Learning
2024	Fetal	Pose-GuideNet: Automatic Scanning Guidance for Fetal Head Ultrasound from Pose Estimation
2024	Infant / neonatal	Super-Field MRI Synthesis for Infant Brains Enhanced by Dual Channel Latent Diffusion

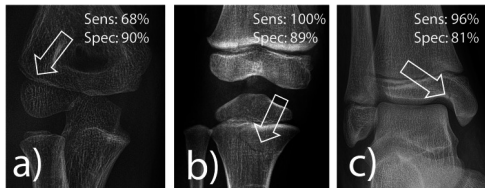
MIDL: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2024	14	–	Imbalance-aware loss functions improve medical image classification
2023	78	–	Patched Diffusion Models for Unsupervised Anomaly Detection in Brain MRI
2023	39	–	MMCFomer: Missing Modality Compensation Transformer for Brain Tumor Segmentation
2023	29	–	Medical diffusion on a budget: textual inversion for medical image generation
2023	23	–	MProtoNet: A Case-Based Interpretable Model for Brain Tumor Classification with 3D Multi-parametric Magnetic Resonance Imaging
2023	22	–	Diffusion Models for Contrast Harmonization of Magnetic Resonance Images
2023	20	–	Exploring Image Augmentations for Siamese Representation Learning with Chest X-Rays
2023	19	–	3D Medical Axial Transformer: A Lightweight Transformer Model for 3D Brain Tumor Segmentation
2023	19	–	Vision-Language Modelling For Radiological Imaging and Reports In The Low Data Regime
2023	14	–	$E(3) \times SO(3)$ -Equivariant Networks for Spherical Deconvolution in Diffusion MRI
2023	14	–	Amortized Normalizing Flows for Transcranial Ultrasound with Uncertainty Quantification
2023	13	–	Selective experience replay compression using coresets for lifelong deep reinforcement learning in medical imaging

MIDL: pediatric radiology AI works, 2023–2026

Year	Population	Paper
2026	Fetal	Calibration-Aware Semi-Supervised Fetal Head Segmentation with Boundary-Positive Contrast
2026	Fetal	Orientation-Aware Diffusion Super-Resolution for 3T-Like Fetal MRI from Routine 1.5T Scans
2026	Fetal	SuD-CoTAN: Sulcal Depth-guided Anatomically Consistent Fetal Cortical Surface Reconstruction
2026	Fetal	Ultra-ECP: Ellipse-Constrained and Point-Robust Foundation Model Adaptation for Fetal Cardiac Ultrasound Segmentation
2026	Pediatric	WristMIR: Coarse-to-Fine Region-Aware Retrieval of Pediatric Wrist Radiographs with Radiology Report-Driven Learning
2025	Fetal	FERN: A Fetal Echocardiography Registration Network for 2D-to-3D Alignment
2025	Pediatric	GAMBAS: Generalised-Hilbert Mamba for Super-resolution of Paediatric Ultra-Low-Field MRI
2024	Fetal	Joint Motion Estimation with Geometric Deformation Correction for Fetal Echo Planar Images Via Deep Learning
2023	Fetal	FUSQA: Fetal Ultrasound Segmentation Quality Assessment

AI fracture detection in a real pediatric emergency department

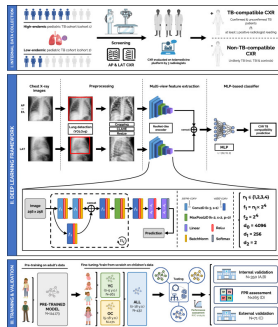


Ziegner et al., Eur Radiol 2025, figure 4: pediatric fractures the software found, with its per-region sensitivity and specificity

source: cdn.ncbi.nlm.nih.gov

- **Problem:** fractures in children are missed most often by the least experienced readers, at night, in the emergency department.
- **Data:** 1,672 consecutive radiographs of children under 18 from one tertiary pediatric emergency department, plus a selected medicolegal set (proximal tibia, radial condyle); three pediatric residents read each before and after seeing the AI.
- **Method:** a commercially available, CE-marked deep-learning fracture detector (adult-derived, pediatric extension) run stand-alone and as a second reader.
- **Result:** stand-alone sensitivity 92%, specificity 83%; 100% for proximal tibia but 68% for radial condyle fractures; residents' accuracy rose from 88% to 90%, and in 2% of cases they dropped a correct diagnosis after seeing the AI.
- **Why it matters:** the first real-life pediatric ED evaluation: the gain from a cleared adult tool is real but modest, and the elbow remains the weak spot.

pTBLightNet: pediatric tuberculosis on frontal and lateral chest radiographs



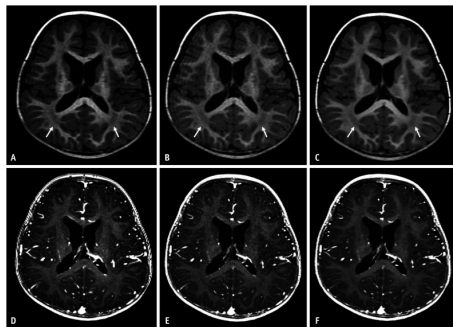
Capellan-Martin et al., Nat Commun 2025, figure 1: the multi-view

pTBLightNet pipeline

source: cdn.ncbi.nlm.nih.gov

- **Problem:** pediatric pulmonary TB has vaguer symptoms and subtler radiographs than adult TB; most affected children in high-burden settings are never diagnosed.
- **Data:** pre-trained on 114,173 adult chest radiographs, then fine-tuned and tested on 918 radiographs from three pediatric TB cohorts, with lateral views.
- **Method:** a multi-view network that scores frontal and lateral films for TB-compatible findings; separate models for under-5 and 5-18 years.
- **Result:** AUC 0.90 on internal and 0.68 on external testing; lateral views helped most in the youngest children; age-specific models matched larger undifferentiated ones.
- **Why it matters:** shows both the value of adult pre-training for a rare pediatric target and the internal-to-external drop that every pediatric deployment should expect.

Deep-learning reconstruction shortens pediatric brain MRI



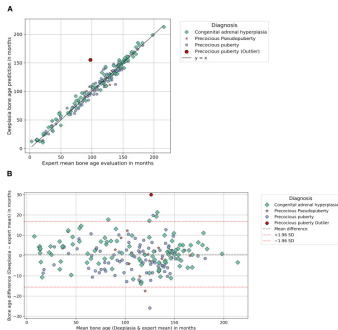
Yoo et al., Korean J Radiol 2025, figure 2: conventional (A, D) vs accelerated 3D T1 pediatric brain MRI without (B, E) and with (C, F)

deep-learning reconstruction

source: cdn.ncbi.nlm.nih.gov

- **Problem:** long 3D T1 sequences are where children move, sedation is prescribed, and scanner time is lost.
- **Data:** 46 children scanned with both the conventional and an accelerated 3D T1 protocol (pre- and post-contrast) on one 3T scanner.
- **Method:** vendor deep-learning reconstruction applied to the accelerated acquisition; noise, contrast, signal-to-noise and radiologist ratings compared with the conventional scan.
- **Result:** acquisition time cut by 29% (pre-contrast) and 41% (post-contrast); the deep-learning images had lower noise, higher SNR and CNR, better overall quality and fewer artifacts than the conventional scan.
- **Why it matters:** reconstruction AI is on the scanner today and gives children a shorter, better exam without any diagnostic model in the loop.

Deeplasia bone age in rare growth disorders, against multiple experts

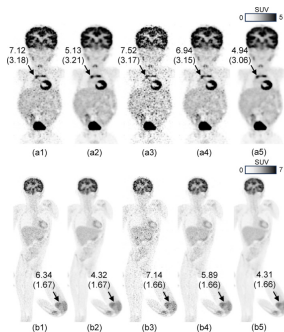


Skaf et al., Front Endocrinol 2026, figure 1: Deeplasia bone age vs the expert mean in the endocrine cohort (scatter and Bland-Altman)

source: cdn.ncbi.nlm.nih.gov

- **Problem:** bone-age models are trained on ordinary hands; the children who need bone age most have syndromes, endocrine disease or lysosomal storage disorders that distort the skeleton.
- **Data:** 1,138 hand radiographs from several centers: SHOX deficiency, Noonan, Silver-Russell, Turner, CAH, precocious puberty (cohort 1) and mucopolysaccharidoses and other storage disorders (cohort 2); two to five expert Greulich-Pyle readings each.
- **Method:** the open-source Deeplasia system run unchanged; each human rater and the model scored against the remaining experts.
- **Result:** mean absolute error 6.0 months (cohort 1) and 7.1 months (storage disorders), 1-year accuracy 90% and 81%; the model beat every individual human rater.
- **Why it matters:** an open pediatric model validated on the hardest cases, by an outside group, on external data: the template for how a children's hospital should test any tool.

Pediatric PET at one fifth of the counts with patch-based deep learning



Han et al., EJNMMI Phys 2026, figure 5: pediatric whole-body FDG

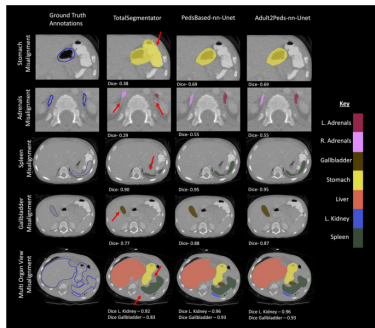
PET, standard 90 s vs 20 s per bed, with and without patch-based

deep-learning denoising

source: cdn.ncbi.nlm.nih.gov

- **Problem:** PET dose and scan time in children are set by noise; conventional denoising networks need large training sets that pediatric PET does not have.
- **Data:** one high-quality pediatric exam for training; a NEMA phantom and 88 clinical whole-body FDG exams (under 12 years) with list-mode data truncated from 90 s to 20 s per bed.
- **Method:** a patch-based network that learns local structure from a single exam and denoises the reduced-count images; blinded observer study and SUV analysis.
- **Result:** 20-second images rated equal or better than the standard 90-second images in more than 85% of cases, with resolution, contrast and SUV preserved.
- **Why it matters:** either a 4.5x shorter scan or about 22% of today's injected activity, from a method trainable on a single local exam.

Children are not small adults: an adult CT segmentation model in children



Chatterjee et al., J Imaging Inform Med 2025, figure 7: pediatric CT

organ labels, ground truth vs adult-trained TotalSegmentator vs

pediatric-trained models

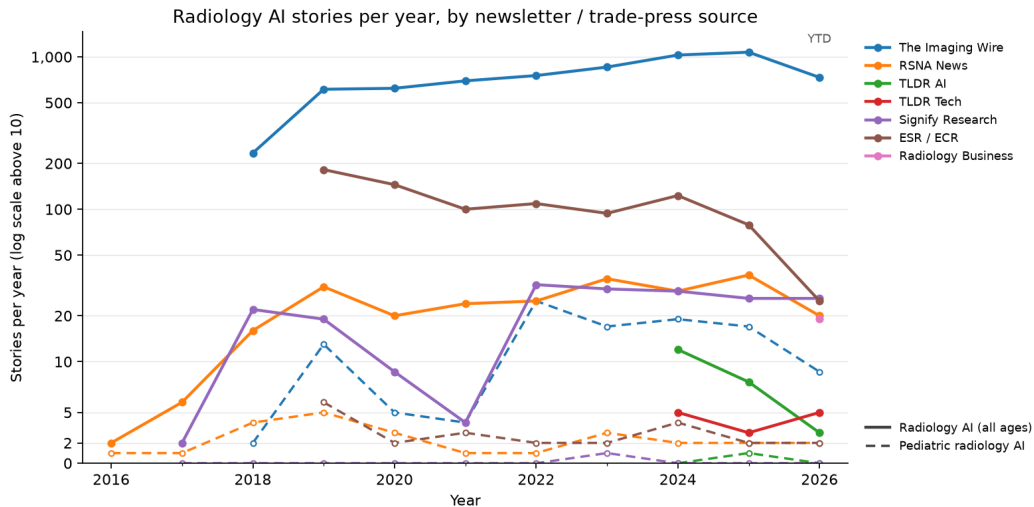
source: cdn.ncbi.nlm.nih.gov

- **Problem:** the most-used open CT organ segmentation model (TotalSegmentator) was trained on adults; nobody had measured how it behaves in children.
- **Data:** external adult ($n = 300$) and pediatric ($n = 359$) abdominal CT sets with expert organ labels.
- **Method:** TotalSegmentator scored on both sets; then a pediatric-only nnU-Net and an adult model fine-tuned on pediatric cases.
- **Result:** mean Dice fell from 0.81 in adults to 0.73 in children, worst for the adrenals (0.69 to 0.41 and 0.35), duodenum and pancreas; pediatric training or fine-tuning recovered the loss.
- **Why it matters:** the clearest quantitative case for local pediatric validation and fine-tuning of adult-derived tools before clinical use.

The biggest public pediatric radiology AI datasets

Dataset	Year	Modality	Size	Ages	Access
CBTN clinical MRI (Children's Brain Tumor Network)	2023	brain / spine tumor MRI	23,101 exams (1,526 patients); tumor masks for 370	pediatric	application (NCI CCDI)
GRAZPEDWRI-DX	2022	wrist radiographs (trauma)	20,327 images (6,091 patients)	0.2–19 y	open
RSNA Pediatric Bone Age Challenge	2017	hand radiographs	14,236 images (bone age + sex)	mean 10.6 y	open (RSNA terms)
FETAL_PLANES_DB	2020	fetal ultrasound (standard planes)	12,400 images (1,792 pregnancies)	fetal	open
ABCD Study	2018–	brain MRI, longitudinal	11,878 participants, up to 7 follow-ups	9–10 y at baseline	application (NBDC)
PediCXR / VinDr-PCXR	2023	chest radiographs	9,125 studies; 36 findings, 15 diagnoses	< 10 y	credentialed (PhysioNet)
Guangzhou pediatric chest X-ray (Kermany)	2018	chest radiographs	5,856 images (normal / bacterial / viral pneumonia)	1–5 y	open
ABIDE I + II (autism)	2012 / 2016	brain MRI (structural, resting fMRI)	2,156 subjects	5–64 y (mostly children)	open
Regensburg Pediatric Appendicitis	2023	abdominal ultrasound + labs	1,709 images (579 patients)	0–18 y	open
dHCP (Developing Human Connectome Project)	2022–24	neonatal + fetal brain MRI	886 neonatal + 297 fetal datasets, with segmentations	20–45 weeks	open (NDA)
Pediatric-CT-SEG (TCIA)	2022	chest / abdomen / pelvis CT	718 series (359 patients); 29 organ contours	5 days – 16 y	open
BraTS-PEDs (pediatric brain tumor challenge)	2023–25	brain tumor MRI	464 patients (2024 challenge); 1,649 series on TCIA	pediatric	registration / TCIA
FeTA (fetal brain tissue annotation)	2021–24	fetal brain MRI	280 volumes (120 train + 160 test); 7 tissue labels	18–35 weeks	registration

Radiology AI newsletter stories per year, by source



Pediatric radiology AI in the trade press, 2023 (23 stories)

Date	Newsletter	Story	Paper
2023-01-08	The Imaging Wire	Gestational Age AI	
2023-03-01	The Imaging Wire	Fetal Ultrasound AI Generalizability	Sendra-Balcells et al., 2023 (Scientific Reports)
2023-03-08	ESR / ECR	ECR 2023 – A true celebration of the cycle of life.	
2023-03-17	The Imaging Wire	When Sao Paolo's Diagnosticos da America (DASA, the world's fourth largest diagn	
2023-04-07	The Imaging Wire	Ultrasound Spots Breech Pregnancies	Knights et al., 2023 (PLOS Medicine)
2023-04-07	The Imaging Wire	When Sao Paolo's Diagnosticos da America SA (DASA, the world's fourth largest di	
2023-04-12	Signify Research	Where do opportunities exist?	
2023-04-12	RSNA News	RSNA Webinar to Examine Impact of Large Language Models	
2023-05-07	The Imaging Wire	New C-Arm Launched	
2023-06-07	The Imaging Wire	BrightHeart's Fetal Heart AI	
2023-06-12	RSNA News	Understanding the Imaging Spectrum of non-Hodgkin Lymphoma in Children, Adolescents and Young Adults	
2023-06-19	The Imaging Wire	Better Together at SIIM	
2023-06-25	The Imaging Wire	AI Detects Child Abuse on CT	
2023-07-07	RSNA News	Survivors of Childhood Cancer: Risk of Future Disease, Screening Needs and the Role of Radiology	
2023-08-02	The Imaging Wire	MRI Radiomics Hones in on Crohn's	Liu et al., 2024 (American Journal of Roentgenology)
2023-08-02	The Imaging Wire	Sonio Gets FDA Nod for Fetal AI	
2023-09-20	The Imaging Wire	GE Lands \$44M AI Ultrasound Grant	
2023-10-08	The Imaging Wire	ChatGPT Reports for Patients	Jeblick et al., 2023 (European Radiology)
2023-11-08	The Imaging Wire	Philips to Expand AI-Powered POCUS	
2023-11-08	The Imaging Wire	Stories of Success in Cloud and AI	
2023-11-12	The Imaging Wire	Uneven Success Against Breast Cancer Pediatric Radiation	
2023-11-26	The Imaging Wire	AI of MRI Diagnoses Autism	
2023-12-11	ESR / ECR	Revolutionising paediatric radiology: the future impact of AI	

Pediatric radiology AI in the trade press, 2024 (25 stories)

Date	Newsletter	Story	Paper
2024-01-10	The Imaging Wire	AI Models Go Head-to-Head in Project AIR Study	van Leeuwen et al., 2024 (Radiology)
2024-01-15	ESR / ECR	Hidden health risks of endocrine disruptors?	
2024-01-17	The Imaging Wire	Echo AI's RHD Impact	Brown et al., 2024 (Journal of the American Heart Association)
2024-01-20	The Imaging Wire	Us2.ai	
2024-02-07	The Imaging Wire	Meet Gleamer at ECR 2024	
2024-03-20	The Imaging Wire	AI Helps Residents Detect Fractures	
2024-04-22	The Imaging Wire	GE Launches US Scanners	
2024-05-19	The Imaging Wire	Early Preeclampsia Risk Screening	Guerby et al., 2024 (Hypertension)
2024-06-29	The Imaging Wire	FDA Clears New Clarius Prenatal App	
2024-07-01	ESR / ECR	AI applications in musculoskeletal imaging	
2024-07-10	RSNA News	Experts Outline Considerations to Deploy AI in Radiology	
2024-08-14	The Imaging Wire	Cardiac Ultrasound Scanner Cleared	
2024-08-25	The Imaging Wire	When Follow-Up Falls Short Pediatric CT Use	
2024-08-25	The Imaging Wire	MAUI Emerges from Stealth	
2024-09-04	The Imaging Wire	Samsung Finalizes Sonio Acquisition	
2024-09-08	The Imaging Wire	AZmed AI Extended to Kids	
2024-10-03	ESR / ECR	On Artificial Intelligence: An interview with Susan Shelmerdine	
2024-10-06	The Imaging Wire	Fracture AI Improves Over Time	
2024-10-10	The Imaging Wire	Low-Dose CT in Kids Hinton Nabs Nobel, Cancels MRI	
2024-10-10	The Imaging Wire	Low-Dose CT Confounds CAD in Kids	Hardie et al., 2025 (American Journal of Roentgenology)
2024-10-17	ESR / ECR	On Artificial Intelligence: An interview with Brendan Kelly	
2024-11-19	RSNA News	Incorrect AI Advice Influences Diagnostic Decisions	
2024-11-25	The Imaging Wire	Fetal Ultrasound AI Cleared	
2024-12-02	The Imaging Wire	Milvue Joins deepc AI Platform	
2024-12-15	The Imaging Wire	Researchers Study AI for Pediatric Fractures	

Pediatric radiology AI in the trade press, 2025 (22 stories)

Date	Newsletter	Story	Paper
2025-01-08	The Imaging Wire	AI for Pediatric Fracture Detection	
2025-01-08	The Imaging Wire	Children's National taps Microsoft to prototype AI tools .	
2025-01-29	The Imaging Wire	Samsung Launches OB/GYN Ultrasound	
2025-02-02	The Imaging Wire	Ultrasound AI Detects Fetal Defects	
2025-02-06	ESR / ECR	Philips at #ECR2025: New AI-enabled systems, intelligent software, and imaging cloud services streamline radiology workflows, advance clinical insights	
2025-02-14	ESR / ECR	Learn more about FUJIFILM's latest innovations at ECR!	Iwan et al., 2020 (European Radiology)
2025-02-27	The Imaging Wire	AI Detects Pediatric Epilepsy	
2025-03-04	TLDR AI	AI to diagnose invisible brain abnormalities in children with epilepsy (4 minute read)	
2025-04-02	The Imaging Wire	AZmed Gets New Chest AI Clearances	
2025-04-13	The Imaging Wire	GE's Cincinnati Pediatric Partnership	
2025-05-14	The Imaging Wire	Philips Partners with NVIDIA on AI for MRI	
2025-05-18	The Imaging Wire	New GE Clearances	
2025-05-20	RSNA News	Radiologists Share Tips to Prevent AI Bias	
2025-06-01	The Imaging Wire	BrightHeart's 3rd FDA Clearance	
2025-07-03	The Imaging Wire	DeepEcho Gets FDA Nod for Fetal AI	
2025-07-10	RSNA News	How Is AI Being Used in Daily Neuroradiology Practice?	
2025-08-03	The Imaging Wire	Radiologist Pay Jumps Nearly 8% in New Survey	
2025-08-13	The Imaging Wire	AI-Assisted Fracture Detection	
2025-08-27	The Imaging Wire	Sonio Launches New Maternal-Fetal AI Options	
2025-10-01	The Imaging Wire	Carebot Expands AI Portfolio	
2025-10-08	The Imaging Wire	Reducing CT Radiation Dose System-Wide	
2025-11-09	The Imaging Wire	AI Detects Pediatric Elbow Fractures	

Pediatric radiology AI in the trade press, 2026 (year to date) (13 stories)

Date	Newsletter	Story	Paper
2026-01-07	The Imaging Wire	BrightHeart Raises EUR 11M	
2026-01-28	The Imaging Wire	AHA Publishes New Stroke Guidelines	Prabhakaran et al., 2026 (Stroke)
2026-01-28	ESR / ECR	Guerbet: Innovation and French Cooperation in the Spotlight at ECR 2026	
2026-02-01	The Imaging Wire	DL-Based CT Reduces Radiation Dose	
2026-02-08	The Imaging Wire	GE Bolsters Ultrasound with Fetal AI	
2026-02-26	ESR / ECR	MRI Unveiling at the European Congress of Radiology (ECR) 2026	
2026-03-29	The Imaging Wire	Ultrasound AI Gets Breakthrough Nod	
2026-05-10	The Imaging Wire	Pediatric MRI Safety, RTs and Remote Radiology, and PSMA-PET	
2026-06-22	RSNA News	Researchers Use AI to Establish Body Charts for CT Imaging Across the Adult Lifespan	
2026-06-28	The Imaging Wire	BrightHeart's Bright Future	Zelop et al., 2025 (Obstetrics & Gynecology)
2026-07-12	The Imaging Wire	AI's Rise in Pediatric Imaging	Shah et al., 2026 (European Radiology)
2026-08-16	The Imaging Wire	CT vs. MRI for Pediatric TBI, Helium Vulnerability, and Ditching the Disk	
2026-08-25	RSNA News	AI Challenges Fuel Innovation Beyond the Leaderboard	

Story 1: a breakthrough designation is not a clearance

The Imaging Wire, 29 March 2026: “Ultrasound AI Gets Breakthrough Nod” (theimagingwire.com)

- **What happened:** the FDA granted **Breakthrough Device Designation** to DeepEcho’s Blind Sweep Platform, a prenatal ultrasound AI that talks the operator through a set of blind sweeps and then reads them.
- **What it does:** from B-mode sweeps rather than expert-acquired standard planes, it automates fetal presentation, placental location and gestational age — the measurements that decide whether a pregnancy needs referral.
- **Why the acquisition end matters:** most fetal AI assumes a sonographer who can already find the view. Moving the AI into acquisition is what makes obstetric ultrasound possible where there is no sonographer, which is where most undetected pathology is.
- **Read the regulatory word carefully:** Breakthrough Designation buys priority FDA interaction, not market authorization. DeepEcho is still working toward clearance, following a separate prenatal algorithm cleared in 2025.

Take-home: trade-press headlines blur designation, clearance and CE mark. For a children’s hospital, only clearance with a pediatric or fetal indication is deployable today.

Story 2: fetal cardiac AI reaches the clinic

The Imaging Wire, 28 June 2026: “BrightHeart’s Bright Future” (theimagingwire.com)

- **What happened:** BrightHeart received the CE mark and debuted its fetal ultrasound AI in the EU at FMF 2026. It now holds **five FDA clearances**, raised a EUR 11M Series A in January 2026, and is installed at Mount Sinai.
- **The evidence behind it:** Zelop et al., *Obstetrics & Gynecology* 2025 — 877 second-trimester examinations from 11 centers, read for eight morphologic findings of severe congenital heart disease; **sensitivity 0.97** for severe CHD, and 0.99 sensitivity / 0.98 specificity against fetal cardiologists for any suspicious finding.
- **The gap it targets:** up to half of major congenital anomalies still go undetected antenatally, and prenatal detection of CHD changes delivery planning and neonatal survival.
- **What is still missing:** the study is retrospective, not prospective. Nobody has yet shown detection rates rise in routine screening practice.

Take-home: this is the pediatric AI story furthest along the path from paper to product — and it is fetal, not pediatric. The pattern to watch is peer-reviewed multicenter accuracy first, clearances second, funding third.

Story 3: what pediatric departments actually report using

The Imaging Wire, 12 July 2026: “AI’s Rise in Pediatric Imaging” — Shah et al., *European Radiology* 2026 (doi 10.1007/s00330-026-12728-9)

- **Why it was news:** after a decade of radiology AI research, almost no data existed on what pediatric departments have actually deployed. This international survey of 18 academic pediatric radiology centers (69% response) is the first real answer.
- **Adoption is wide but shallow:** **88.9%** run at least one AI tool, yet the tools cluster on two tasks — bone age (44.4%) and segmentation / quantification (22.2%) — and only **16.7%** called AI “transformational”.
- **The barriers named:** lack of pediatric datasets (83.3%), integration into PACS and workflow (66.7%), cost (50%).
- **It matches this deck’s own data:** bone age is also the most common pediatric clinical problem in the published literature and the one task with a mature commercial product.

Take-home: the constraint is not model accuracy. It is pediatric training data and integration — both of which a children’s hospital, not a vendor, is positioned to fix.

Story 4: challenges as pediatric data infrastructure

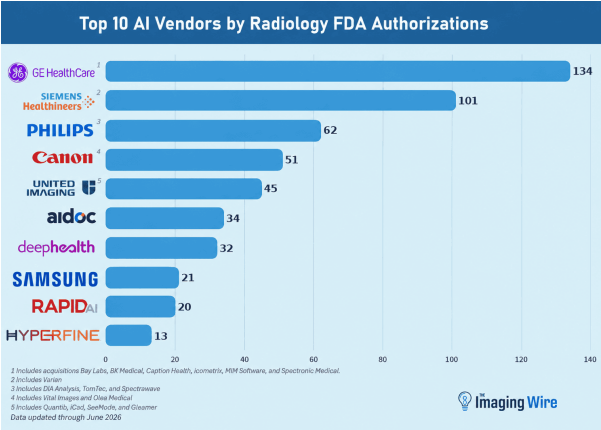
RSNA News, 25 August 2026: “AI Challenges Fuel Innovation Beyond the Leaderboard” (rsna.org)

- **What happened:** RSNA looked back at a decade of its Kaggle AI Challenges and argued that the lasting product is not the winning model but the **curated multinational datasets with expert annotations**, which stay public after the leaderboard closes, plus the open-sourced top solutions.
- **The pediatric one:** the **2017 Pediatric Bone Age Challenge** drew 700+ entries and remains one of the largest open pediatric imaging datasets (14,236 hand radiographs).
- **The generalization lesson:** Kitamura’s team placed third, then revalidated on Brazilian data and found performance did not transfer unchanged — the same population-shift problem that makes adult-trained tools unsafe in children.
- **The honest caveat:** Pan notes these algorithms target one urgent finding and behave as triage, not as a general assistant — leaderboard rank does not predict clinical usefulness.

Take-home: challenges are how pediatric imaging has historically gotten public data. Contributing annotated pediatric cases is a concrete, low-cost way for a children’s hospital to move the field.

Radiology keeps its lead in FDA AI authorizations

The Imaging Wire, 14 September 2026: FDA AI-enabled device list, updated through June 2026



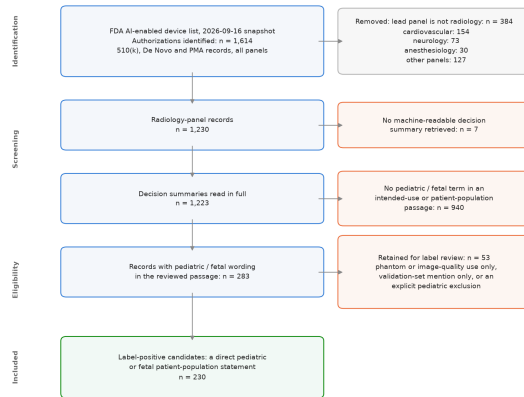
- **1,614** AI-enabled devices authorized since the FDA began tracking, up 5.9% from Q1 2026.
- **1,230** of them (76%) are radiology, the same radiology-panel total this deck's own FDA pull finds.
- Q2 2026: 89 authorizations (92 in Q1), 66 of them (74%) radiology.
- Scanner makers lead: GE HealthCare 134, Siemens Healthineers 101, Philips 62, Canon 51, United Imaging 45.
- The largest software-only vendors are Aidoc (34), DeepHealth (32) and RapidAI (20).
- The FDA is still deciding how to regulate generative AI used in direct clinical care.

Top 10 vendors by radiology FDA authorizations, counts include acquisitions

source: theimagingwire.com

How many cleared devices actually name children?

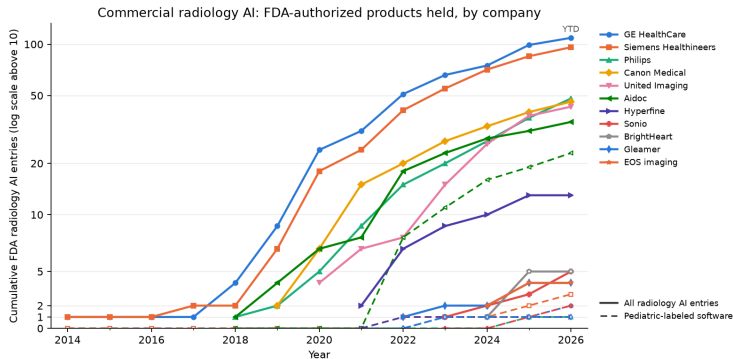
FDA AI-enabled device list: which authorizations name children



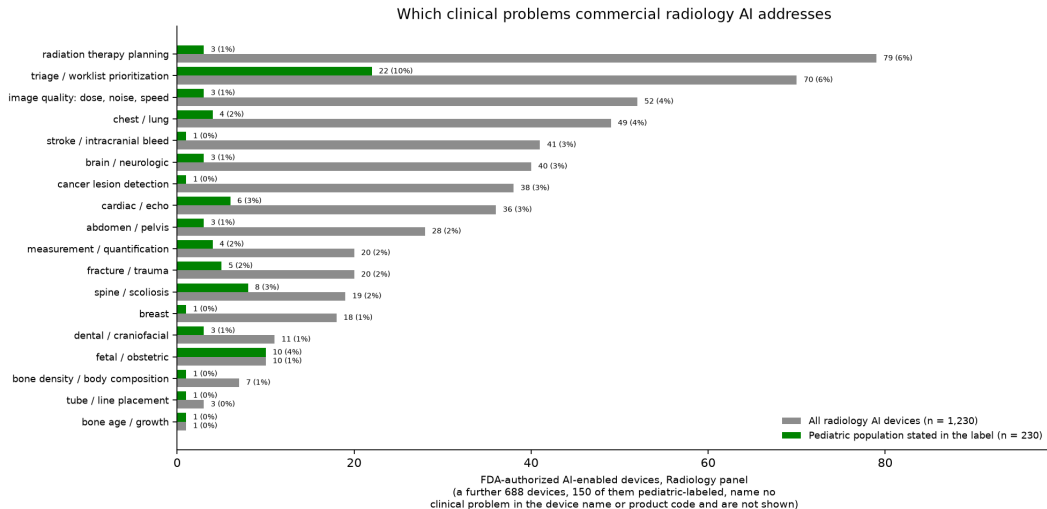
Keyword screen of the linked FDA decision summaries, not a determination of pediatric authorization:
confirm the current labeling before purchase. One row per submission, so versions and resubmissions count separately.

230 of the 1,230 radiology-panel authorizations (19%) name a pediatric or fetal population in the labeling passage reviewed; 53 more name children only as a phantom, a validation set or an exclusion, and are kept for label review.

Who is building commercial radiology AI



Which clinical problems the cleared products address



Products a pediatric radiologist should know (part 1)

Product	Vendor	Clinical problem	What it does	Pediatric scope	Where it stands
BoneXpert	Visiana	bone age / growth	Automated Greulich–Pyle and Tanner–Whitehouse bone age plus a Bone Health Index, from a hand radiograph	Built for children; CE-marked and in routine European use; US research use only	Widely deployed (Europe)
BoneView	Gleamer	fracture / trauma	Fracture detection on trauma radiographs, presented as regions of interest alongside the image	US indication covers ages 2–21 for specified extremity views; pelvis, hip, femur, ribs and spine are adult-only [K222176, 2023]	Widely deployed
Rayvolve (AZtrauma)	AZmed	fracture / trauma	Fracture and MSK finding detection on radiographs; sibling modules cover chest findings, measurements and bone age	US fracture indication from age 2; the bone-age module is CE-marked only [K240845, 2024]	Widely deployed
TechCare Trauma	Milvue	fracture / trauma	Trauma radiograph triage: fractures, effusions and dislocations flagged for the reader	The decision summary states a pediatric population [K242171, 2025]	Early US deployments
EFAI Bonesuite XR Bone Age Pro	Ever Fortune.AI	bone age / growth	Greulich–Pyle bone age from a PA left hand and wrist radiograph, as an adjunct to the radiologist	FDA-cleared for ages 2–16 — the pediatric-specific US bone-age clearance [K234042, 2024]	FDA-cleared, early adoption
Physis	16 Bit	bone age / growth	Bone-age estimation with Greulich–Pyle atlas matching and draft report text	Pediatric tool; Health Canada licensed, no FDA authorization identified	Promising, limited US access
Fetal EchoScan	BrightHeart	fetal / obstetric	Flags eight morphological findings suspicious for congenital heart disease on second-trimester views	Prenatal by definition: second-trimester exams, maternal age 18+ [K252294, 2025]	Early deployments / pilots
Sonio Detect / Sonio Suspect	Sonio	fetal / obstetric	Checks that the required fetal views were obtained and are of usable quality (Detect); flags images suspicious for anomalies (Suspect)	Prenatal by definition; every FDA entry states a fetal population [K252433, 2026]	Early deployments / pilots
Delivery Date AI	Ultrasound AI	fetal / obstetric	Predicts the delivery date from a second- or third-trimester ultrasound when dating is unreliable	Prenatal; the first De Novo of its kind (a new device class) [DEN250007, 2026]	Newly authorized, unproven in practice
Swoop Portable MR + BrainInsight	Hyperline	brain / neurologic	Bedside low-field brain MRI whose usable image quality comes from deep-learning reconstruction; BrainInsight adds automated volumes and midline shift	Decision summaries state a pediatric population; the natural use is the NICU and PICU, where transport is the risk [K253489, 2025]	In pediatric pilots
spineEOS / VEA Align	EOS imaging (Alphatec)	spine / scoliosis	Automated 3D spine and lower-limb modeling from biplanar low-dose full-body radiographs	Scoliosis follow-up is repeated imaging of a growing child, so the dose saving compounds [K251747, 2025]	Widely deployed (pediatric orthopedics)
CoLumbo	Smart Soft Healthcare	spine / scoliosis	Spine MRI segmentation, measurement and structured reporting, lumbar and cervical	Decision summaries state a pediatric population; no pediatric-specific performance reported [K254015, 2026]	Early deployments

Square brackets give the FDA submission supporting the pediatric claim and its decision year; a blank means no US authorization was identified for that product. “Where it stands” is our reading of deployment, not a measured adoption rate. A company’s total of FDA AI entries is not a pediatric credential — the large vendors’ entries are mostly whole scanners whose labels list pediatric imaging as one clinical application (see the authorizations-per-company figure).

Products a pediatric radiologist should know (part 2)

Product	Vendor	Clinical problem	What it does	Pediatric scope	Where it stands
BriefCase-Triage (CARE modules)	Aidoc	triage / worklist prioritization	Worklist triage on CT and CTA: intracranial hemorrhage, pulmonary embolism, pneumothorax, cervical-spine and rib fractures, free gas and more	Most entries state a pediatric population; the evidence behind them is adult, so treat pediatric use as local validation work [K253578, 2026]	Widely deployed
Viz LVO / Viz ICH	Viz.ai	stroke / intracranial bleed	Detects large-vessel occlusion and intracranial hemorrhage and pushes the case to the stroke team's phones	No pediatric population statement found in the screened summaries; pediatric stroke use would be off-label	Widely deployed (adult)
qXR-TB	Qure.ai	chest / lung	Chest radiograph screening for tuberculosis, built for high-burden settings with few readers	Vendor reports CE-marked use from age 0–15; US eligibility varies by module	Deployed in global-health screening
AIR Recon DL / TrueFidelity	GE HealthCare	image quality: dose, noise, speed	Deep-learning reconstruction: shorter MRI acquisitions at the same quality, and CT images at lower dose	Sold across ages; the pediatric benefit is fewer sedations and less dose, and it is the AI most children actually meet	Widely deployed
Deep Resolve	Siemens Healthineers	image quality: dose, noise, speed	MRI deep-learning denoising and resolution enhancement that supports accelerated acquisition	Vendor publishes pediatric protocols; scan-time reduction is the pediatric argument	Widely deployed
AIce	Canon Medical	image quality: dose, noise, speed	Deep-learning reconstruction for both CT (noise, dose) and MRI (noise, resolution, scan time)	Vendor publishes pediatric CT galleries; protocol and version determine what is available	Widely deployed
SmartSpeed / Precise Image	Philips	image quality: dose, noise, speed	MRI acceleration with deep-learning reconstruction (SmartSpeed); CT reconstruction at reduced dose (Precise Image)	Sold across ages; pediatric protocols are the vendor's own, not a separate indication	Widely deployed
SubtleMR / SwiftMR	Subtle Medical / AIRS Medical	image quality: dose, noise, speed	Vendor-neutral MRI post-processing: denoising and sharpening so a shorter acquisition still reads	No adult-only restriction in the reviewed US indications; SwiftMR's 2026 validation includes ages 0–21	Deployed, vendor-neutral retrofit
Butterfly iQ3; Clarius Bladder AI	Butterfly Network / Clarius	point-of-care ultrasound	Handheld ultrasound with on-device AI: bladder volume, gestational age, ejection fraction, lung and nerve tools	The scanners are labeled for adult and pediatric imaging, but each AI tool carries its own age scope and most are adult-only; Clarius Bladder AI covers ages 2 and over [K232257, 2023]	Scanners widely deployed; AI tools mostly adult-scoped
Arterys Cardio DL	Arterys (Tempus)	cardiac / echo	Cardiac MRI segmentation, flow and volume quantification with a cloud workflow	One of the earliest AI clearances to name neonates, infants, children and adolescents outright (2017) [K163253, 2017]	Historically important; now a Tempus product
Acorn 3D	Mighty Oak Medical	spine / scoliosis	Automated vertebral segmentation from CT to build patient-specific 3D models and surgical guides	Decision summaries state a pediatric population; the use case is pediatric deformity surgery [K260322, 2026]	Niche, surgical
Second Opinion Pediatric	Pearl	dental / craniofacial	Detects caries and other findings on pediatric dental radiographs	The rare device whose name and indication are pediatric from the start [K243893, 2025]	Deployed in dentistry

Square brackets give the FDA submission supporting the pediatric claim and its decision year; a blank means no US authorization was identified for that product. “Where it stands” is our reading of deployment, not a measured adoption rate. A company's total of FDA AI entries is not a pediatric credential — the large vendors' entries are mostly whole scanners whose labels list pediatric imaging as one clinical application (see the authorizations-per-company figure).

Kamran review: broader research, stronger validation

Kamran et al., *Pediatric Radiology* 2026 (doi 10.1007/s00247-025-06462-5)

- **A large but concentrated literature:** 789 pediatric-focused original articles (2005–August 2024); 91% addressed image interpretation/diagnosis. Leading applications were bone age (88), pneumonia (65), and brain lesions/tumors (54).
- **Generalizability needs stronger evidence:** 65% relied on local hospital datasets. The authors call for diverse, multicenter datasets and prospective validation across pediatric ages, sites, and disease-prevalence settings.
- **Broaden the research agenda:** communication, education, policy, and stakeholder perspectives were underrepresented. Include children, families, and clinical teams in development and evaluation.

Take-home: prioritize representative data, validation, and neglected clinical needs. This review maps research activity; it does not compare model accuracy or establish clinical readiness.

ACR white paper: the pediatric AI gap

Sammer et al., *J Am Coll Radiol* 2023;20:730–737 (doi 10.1016/j.jacr.2023.06.003)

- **Scope:** ACR Pediatric AI Workgroup white paper framing limited access to pediatric AI as a health equity issue.
- **Availability:** at the time of the review, only 6 of more than 200 FDA-cleared tools in the ACR catalog indicated pediatric use (about 3%).
- **Why adult AI may fail:** growth, disease patterns, and imaging protocols differ; published studies show uneven performance across pediatric subgroups.
- **Barriers:** scarce labeled pediatric data, consent and privacy concerns, and weak financial incentives for smaller pediatric markets.
- **Recommendations:** validate in children, disclose age-specific evidence, fund pediatric development, involve families, and protect pediatric workflows. The authors introduce *Image IntelliGently* to advance this work.

Multisociety statement: safe pediatric AI implementation

Shelmerdine et al., *J Am Coll Radiol* 2026;23:89–101 (doi 10.1016/j.jacr.2025.08.019)

ACR, ESPR, SPR, SLARP, AOSPR, and SPIN propose four pillars for safe adoption:

- **Regulation and purchasing:** pediatric labeling and proposed safety ratings; diverse datasets, subgroup performance, transparency, and cybersecurity.
- **Implementation and integration:** define a local clinical need, select tools validated in children, pilot before scaling, and involve pediatric specialists and families.
- **Interpretation and postmarket surveillance:** retain human oversight; audit performance and patient outcomes; manage discrepancies, report adverse events, and address automation bias.
- **Education:** foundational AI literacy for all staff, pediatric-specific training, continuing education, and patient/public engagement.

Worth knowing as a radiologist

- **Deep Learning: A Primer for Radiologists** (Radiographics paper, 2017): a helpful introduction to the principles and applications of deep learning in radiology; essential for understanding how AI tools work and their limitations
- **RSNA Pediatric Bone Age Challenge (2017)** (benchmark / dataset): a landmark pediatric benchmark; useful for understanding automated bone-age accuracy and its limits
- **Gleamer BoneView / AZmed Rayvolve** (clinical products): examples of fracture-detection assistance with pediatric indications; eligibility depends on age, anatomy, and product version
- **FDA AI-enabled device list and linked labeling** (regulatory resource): a starting point for checking authorized use and pediatric age ranges; the list is not exhaustive
- **CLAIM checklist (2024 update)** (reporting guideline): check whether data, reference standards, and testing are adequately described; reporting completeness alone does not establish clinical benefit
- **TotalSegmentator** (research tool): automated CT / MR organ segmentation for volumes and body composition; check accuracy in children before trusting measurements
- **nnU-Net** (research framework): a strong baseline for training segmentation models; ask whether a new method improves on an appropriate comparator
- **MedSAM / segment-anything models** (research models): prompt-guided contours can assist annotation; radiologists still need to review and correct the output
- **MONAI** (research framework): infrastructure for imaging-AI development; useful when collaborating with a research team, rather than a ready-to-use diagnostic tool

Recommendations for pediatric radiologists

- AI development is constrained by scarce public datasets, fragmented institutional data, rare diseases, and heterogeneous reporting practices.
- Choose one task worth improving in your practice: bone-age measurements, fracture detection, or reducing CT dose / MRI scan time.
- Before adopting a tool, ask which children it was tested on. Check the intended age range and test it on your own cases, including younger children and unusual anatomy.
- In a local pilot, measure what changes with AI: reading time, missed findings, false alarms, dose, or scan time. Review errors and keep checking after rollout.
- Help build the evidence we need: choose a pediatric problem you see often, label cases carefully, and work with other hospitals to test whether a model works beyond the site that built it.